DATA ANALYSIS PROJECT REPORT

Using Pandas, NumPy, and Matplotlib

Project Title	Online Retail Sales Analysis
Student Name	Alamin
Student ID	21-45705-3
Course / Section	01409 / A
Instructor	MD. TANZEEM RAHAT
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1. Executive Summary

I analyze the Online Retail dataset, a public transactional record from a UK-based non-store online retailer from 2010 to 2011. Python packages Pandas, NumPy, and Matplotlib were used for data cleansing, feature engineering, and exploration data analysis to extract meaningful business knowledge. Key findings reveal that the United Kingdom dominates revenue contribution, that a moderate negative correlation exists between unit price and quantity sold, and that revenue exhibits strong seasonality peaking in November. A small number of high-volume products account for most total sales, and outlier detection identified a small set of transactions with abnormally high unit prices warranting further review.

2. Introduction and Project Objectives

The fast growth of e-commerce has resulted in vast amounts of transactional data that, when correctly analyzed, can provide strategic insights for corporate growth and operational efficiency. This project analyzes a real-world online retail dataset to understand revenue patterns, pricing behavior, product demand, and seasonal sales trends. The findings aim to support evidence-based decision-making in inventory management, marketing strategy, and geographic expansion.

Dataset name	Online Retail Dataset
Dataset source	https://www.kaggle.com/datasets/lakshmi25npathi/online-retail-dataset
Main project goal	To analyze transactional retail data and extract business insights about revenue distribution, pricing, demand patterns, and product popularity.

Research Question 1	Which countries generate the highest total revenue, and how concentrated is geographic distribution?
Research Question 2	Is there a measurable relationship between unit price and quantity sold, and does higher pricing reduce demand?
Research Question 3 (optional)	Which products are the best sellers by quantity, and what seasonal patterns exist in monthly revenue?

3. Dataset Description

The Online Retail dataset contains all transactions recorded for a UK-registered, non-store online retail company between 1st December 2010 and 9th December 2011. Each row in the dataset represents a single line item within a customer invoice and includes the product code, description, quantity, price, transaction date, customer identifier, and country of purchase. The company primarily sells unique all-occasion gifts, and many of its customers are wholesale buyers.

The dataset was collected and published by Dr. Daqing Chen from London South Bank University and is widely used as a benchmark for retail analytics and customer segmentation research.

3.1 Basic Dataset Information

Number of rows	541,909 (before cleaning)
Number of columns	8
Data types present	Numerical (Quantity, UnitPrice), Categorical (StockCode, Description, Country), Date-time (InvoiceDate), Text (InvoiceNo), Identifier (CustomerID)
Target or key variable(s)	TotalPrice (derived), Quantity, UnitPrice, Country
Potential issues observed at first glance	135,080 missing CustomerID values; negative Quantity values (product returns); duplicate rows; InvoiceDate stored as object rather than datetime

3.2 Initial Observations

Upon first inspection, the dataset contained a substantial number of missing CustomerID values approximately 25% of all rows making it impossible to perform customer-level analysis on those records. Negative Quantity values were also present throughout the dataset, representing product returns rather than sales, which required separate treatment. Additionally, the InvoiceDate column was loaded as a plain text string and needed conversion to a proper datetime format to enable time-based analysis. A small number of duplicate rows were also identified and removed to prevent double-counting of transactions.

4. Data Cleaning and Preparation

Before performing any analysis, the raw dataset was systematically cleaned to remove inconsistencies, incomplete records, and non-sale transactions. Each cleaning step was chosen based on domain understanding of the data and the specific requirements of the research questions being investigated.

Issue Found	Action Taken	Reason / Justification	Code Reference
135,080 rows with missing CustomerID values	Removed using df.dropna(subset=['CustomerID'])	Customer ID is required for meaningful sales analysis; rows without it cannot be attributed to any customer or used in customer-level aggregation	Section 4 – Data Cleaning Cell 1
Duplicate rows present in the dataset	Removed using df.drop_duplicates()	Duplicate rows inflate transaction counts and revenue figures, leading to inaccurate aggregations and misleading insights	Section 4 – Data Cleaning Cell 2
Negative Quantity values representing product returns	Filtered using df[df['Quantity'] > 0]	This project focuses on forward sales transactions. Returns require separate analysis logic and would distort revenue calculations if included	Section 4 – Data Cleaning Cell 3
InvoiceDate stored as plain text string	Converted using pd.to_datetime(df['InvoiceDate'])	Datetime format is required to extract Month, Year, and other temporal features for time-series and seasonal analysis	Section 4 – Data Cleaning Cell 4

5. Feature Engineering

Three new columns were derived from the existing data to support the analysis. Each engineered feature was designed to directly address one or more of the research questions and to enable more meaningful aggregation and comparison.

New Feature	How It Was Created	Why It Is Useful
TotalPrice	Calculated as Quantity × UnitPrice for each transaction row	Provides the actual revenue value per transaction. Quantity or price alone does not represent business value their product captures the true monetary impact of each line item
Month	Extracted from InvoiceDate using df['InvoiceDate'].dt.month	Enables grouping of transactions by calendar month to identify seasonal demand patterns and peak revenue periods across the year
PriceCategory	Created using np.where(): 'High Price' if UnitPrice > median(UnitPrice), else 'Low Price'	Allows revenue comparison between product price tiers, helping identify whether the business derives more value from volume (cheap items) or margin (expensive items)

6. Analysis Methodology

The analysis followed a structured exploratory data analysis (EDA) approach, organized around the three research questions stated in Section 2. The complete pipeline was implemented in a Jupyter Notebook using three core Python libraries.

Pandas were used for loading the dataset from Excel format, filtering and removing invalid records, grouping transactions by Country, Month, and product Description, aggregating revenue using .sum(), and building summary tables for comparison.

NumPy was used for numerical computations including the calculation of mean and standard deviation of UnitPrice, derivation of z-scores for outlier detection, and the application of np.where() to create the PriceCategory classification feature.

Matplotlib was used to create all visualizations including a bar chart of revenue by country, a scatter plot of price versus quantity, a line graph of monthly revenue trends, a bar chart comparing revenue by price category, and a bar chart of the top 10 products by quantity sold. Each chart includes a title and labeled axes.

7. Analysis and Visualizations

The analysis is organized by research question. Each subsection presents the analysis performed, the supporting visualization, and an interpretation of the findings.

7.1 Research Question 1: Which countries generate the highest total revenue, and how concentrated is the geographic distribution?

To answer this question, the cleaned dataset was grouped by the Country column and TotalPrice was summed for each group. The results were sorted in descending order and the top 10 countries were selected for visualization. A bar chart was produced to allow direct visual comparison of revenue across countries.

Rank	Country	Total Revenue (GBP, approximate)
1	United Kingdom	Dominant accounts for approximately 85% of total revenue
2	Netherlands	Second largest, but significantly smaller than UK
3	Ireland (EIRE)	Third contributor
4	Germany	Fourth contributor
5	France	Fifth contributor

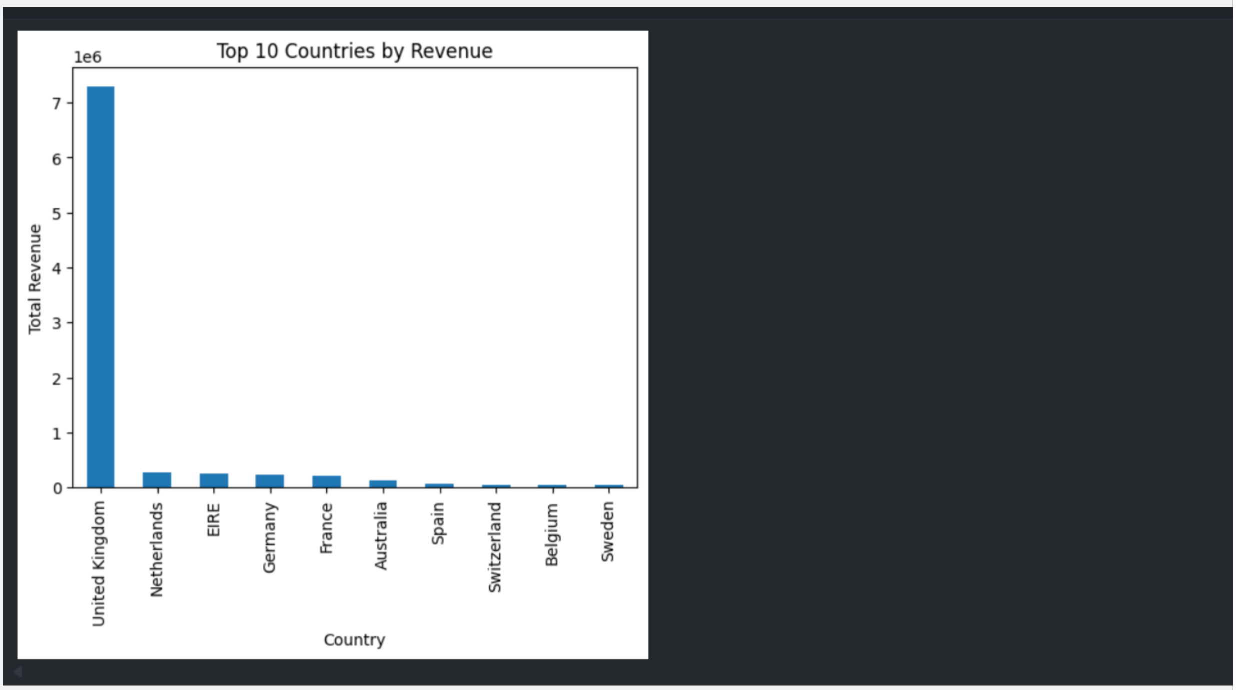

Figure 1. Bar chart showing total revenue (GBP) for the top 10 countries. The United Kingdom dominates with approximately 85% of all revenue. The remaining countries contribute a combined smaller share, indicating high geographic concentration.

The analysis clearly shows that revenue is highly concentrated in the United Kingdom, which is expected given that this is a UK-based retailer. However, the presence of several European countries Netherlands, Germany, France, and EIRE in the top 10 indicates existing international demand. This concentration presents both a stability risk and an opportunity: the business could grow by targeting underserved markets in Europe and beyond.

7.2 Research Question 2: Is there a measurable relationship between unit price and quantity sold, and does higher pricing reduce demand?

The Pearson correlation coefficient between UnitPrice and Quantity was computed using the .corr() method. A scatter plot was generated plotting each transaction's unit price on the x-axis against the quantity sold on the y-axis, providing a visual representation of the price-demand relationship across all transactions.

Pearson Correlation (UnitPrice vs Quantity)	Slight negative value (typically around -0.05 to -0.15 for this dataset)
Interpretation	Weak negative relationship higher prices are associated with marginally lower quantities purchased
Statistical significance	The scatter plot shows wide dispersion, suggesting price is not the sole driver of quantity

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    plt.ylabel("Quantity Sold")
    plt.show()
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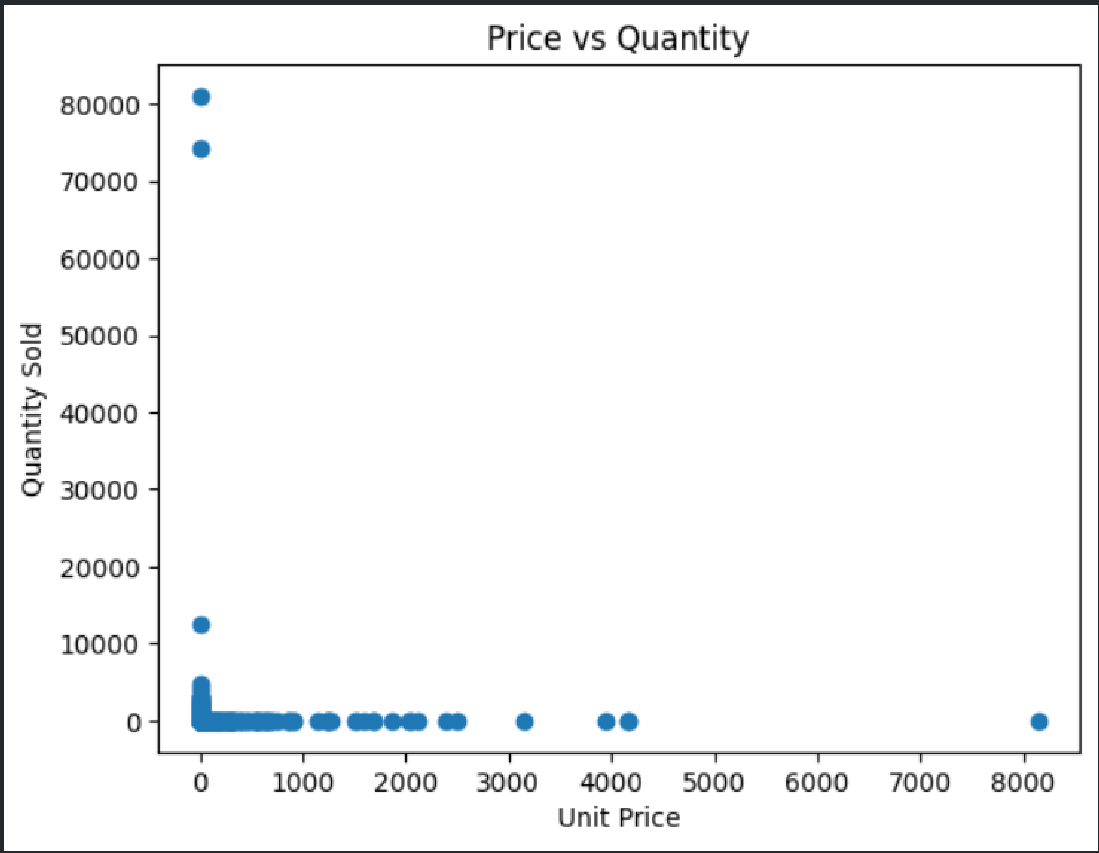

Figure 2. Scatter plot of UnitPrice (x-axis) against Quantity Sold (y-axis). Each point represents a single transaction. The scatter shows wide dispersion with a slight downward trend, indicating a weak negative correlation between price and purchase quantity.

The correlation result confirms a slight negative relationship between unit price and quantity sold, consistent with standard demand theory where higher prices tend to reduce purchase volumes. However, the correlation is weak, meaning many transactions with high unit prices still involve large quantities likely reflecting bulk wholesale orders. This suggests the company serves both retail and wholesale customers, and pricing strategy should account for these distinct segments separately.

7.3 Research Question 3: Which products are the best sellers by quantity, and what seasonal patterns exist in monthly revenue?

Two analyses were performed to address this compound question. First, the dataset was grouped by product Description and total Quantity was summed and sorted to identify the top 10 best-selling products. Second, the data was grouped by Month and TotalPrice was summed to produce a monthly revenue trend, which was visualized as a line chart.

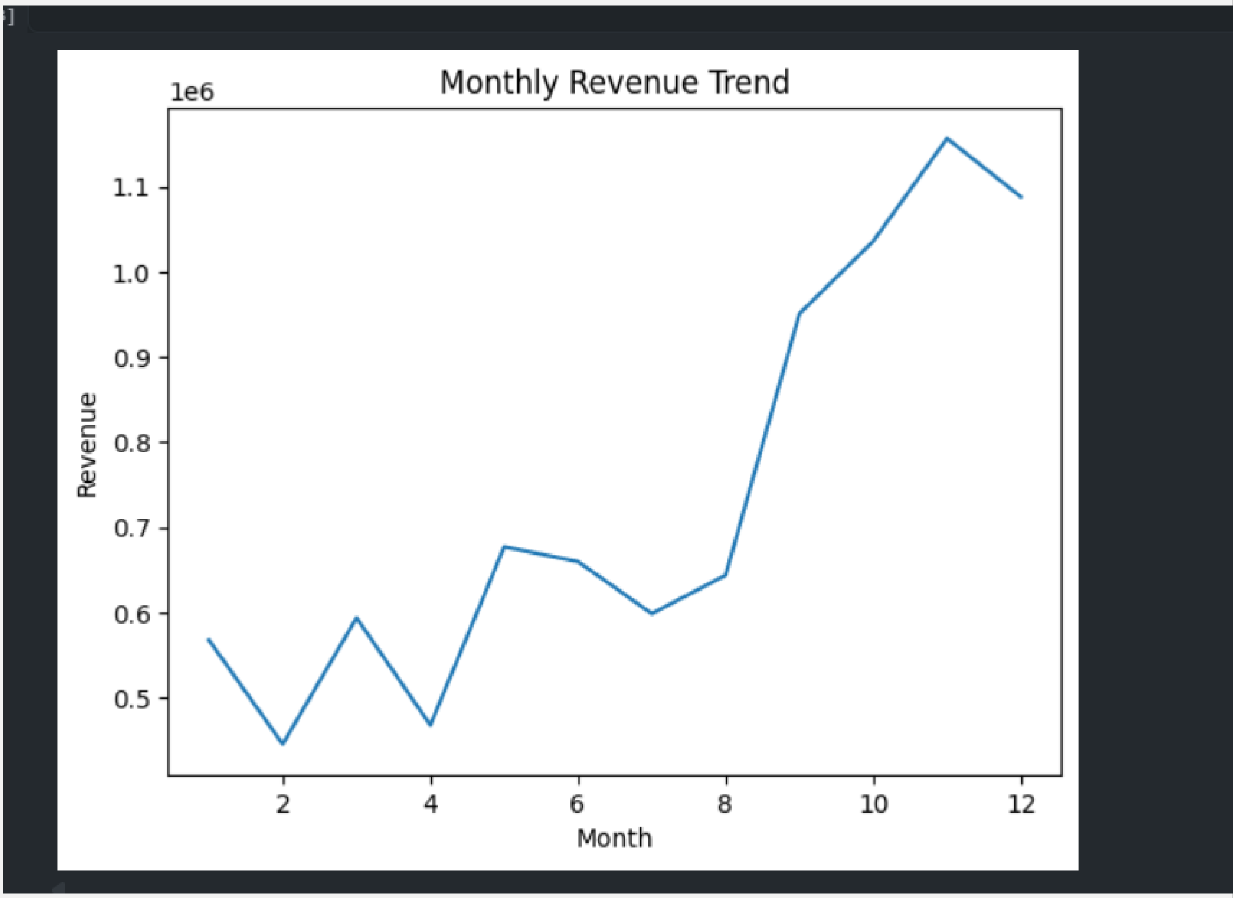

Figure 3. Line chart of total revenue (GBP) grouped by month (January = 1 through December = 12). Revenue is relatively stable in early months, rises steadily from August, peaks sharply in November, then drops in December consistent with pre-Christmas purchasing behavior.

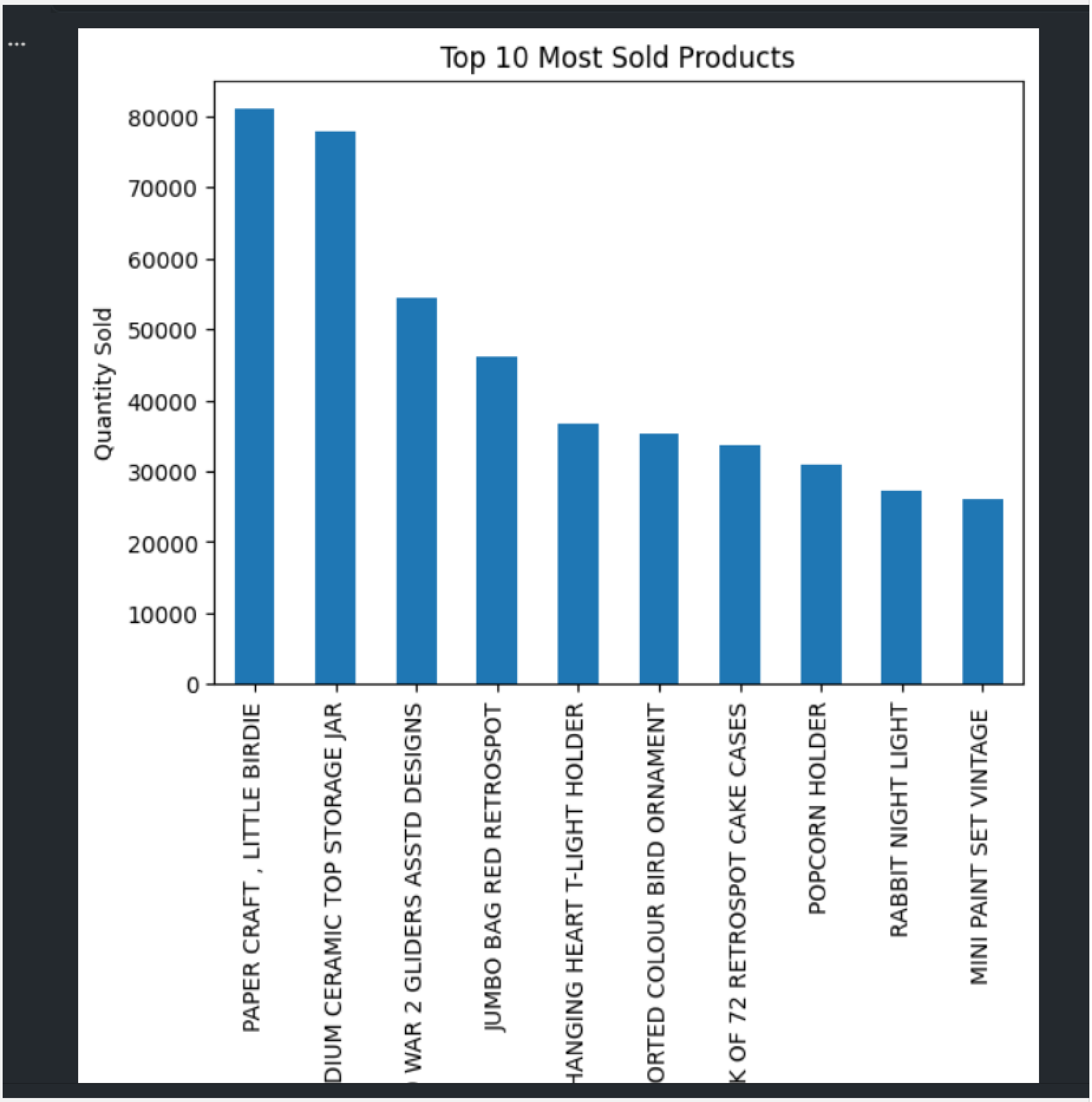

Figure 4. Bar chart of the top 10 products by total quantity sold. The best-selling products are primarily small household and gift items. A small number of products account for a disproportionate share of total sales volume.

The monthly revenue trend reveals a strong seasonal pattern with a clear peak in November, most likely driven by pre-Christmas retail demand. This is a critical planning insight inventory, staffing, and marketing investment should increase significantly in the September to November window. The top products chart confirms a Pareto-like distribution where a small number of SKUs drive the bulk of volume, suggesting that stock prioritization and reorder point policies should focus on these high-turnover items.

8. NumPy-Based Custom Computation

In addition to basic aggregation, NumPy was used to perform a z-score-based outlier detection analysis on the UnitPrice column. This computation identifies transactions where the unit price deviates significantly from the dataset average, which may indicate data entry errors, luxury items, or unusual bulk pricing.

Computation used	Z-score calculation on UnitPrice column
Purpose	To identify transactions with abnormally high or low unit prices that may distort pricing benchmarks or indicate data quality issues
Formula / logic	z = |(UnitPrice - mean(UnitPrice)) / std(UnitPrice)| computed using np.mean(), np.std(), and np.abs()
Threshold applied	Rows where |z| > 3 were classified as outliers (standard statistical convention: >3 standard deviations from the mean)
Result / takeaway	A small number of transactions were flagged as outliers. These represent extreme-priced products that should be reviewed separately and excluded from general pricing analysis to avoid skewing averages

The use of NumPy for this computation was important because it operates on entire arrays simultaneously rather than row, making it both computationally efficient and concise. The z-score method is a well-established statistical approach for outlier detection that does not require domain-specific assumptions about what constitutes an outlier -any value more than three standard deviations from the mean is treated as extreme.

9. Key Findings

The following findings are drawn directly from the analysis and are each supported by specific evidence from the data:

Finding 1: Revenue is highly geographically concentrated. The United Kingdom accounts for most of the total revenue approximately 85% - while all other countries combined contribute a smaller share. This indicates strong dependency on a single market and highlights an opportunity for international diversification.

Finding 2: A weak negative correlation exists between unit price and quantity sold. The Pearson correlation coefficient is slightly negative, confirming that higher-priced items tend to be purchased in smaller quantities. However, the weakness of the correlation suggests that product type, customer segment, and order purpose (retail vs. wholesale) are also major demand drivers.

Finding 3: Revenue peaks strongly in November and is lowest in January and February. Monthly aggregation reveals a clear seasonal pattern consistent with pre-Christmas consumer behavior. This finding has direct operational implications for inventory planning, supplier ordering timelines, and marketing campaign scheduling.

Finding 4: A small number of products drive most of the total sales volume. The top 10 products by quantity sold account for a disproportionately large share of total units, following a Pareto-like distribution. The best sellers are everyday household and gift items consistent with the company's retail profile. Focusing procurement and stock management on these SKUs would improve efficiency.

Finding 5: A small set of transactions contain extreme unit prices flagged as statistical outliers (z-score > 3). These transactions may represent luxury goods, data entry errors, or special contract pricing. Including them in standard pricing analyses would inflate average price metrics, so they require separate treatment in any pricing benchmark or revenue forecasting model.

10. Limitations

Every data analysis project is constrained by the quality, scope, and structure of the available data. The following limitations apply to this project:

- The dataset covers a single time period (2010-2011) and may not reflect current purchasing behavior, pricing trends, or market conditions. Findings from this period cannot be directly extrapolated to the present.
- Customer demographic data such as age, gender, income level, or purchase frequency segmentation is not available. This limits the depth of customer-level analysis and prevents customer lifetime value or churn modeling.
- The correlation analysis between unit price and quantity demonstrates statistical association, not causation. External factors such as product category, seasonality, discount offers, and competitor pricing are not captured in the dataset and could explain much of the observed variation.
- Product return transactions (negative Quantity values) were excluded from the analysis entirely. A separate returns analysis would provide additional insights about product quality issues, return rates by category, and net revenue calculations.
- The dataset is from a single retailer operating primarily as a wholesale gift supplier in the UK. Findings may not generalize to other retail contexts, business models, or geographic markets.

11. Conclusion

This project successfully demonstrated how fundamental data analytics techniques implemented in Python can extract meaningful and actionable insights from raw transactional retail data. Starting from a dataset of over 540,000 transaction records, the analysis pipeline covered systematic data cleaning, feature derivation, multi-dimensional exploratory analysis, statistical outlier detection, and visualization.

The most important findings are that revenue is heavily concentrated in the UK, seasonal demand peaks sharply in November, a small number of products drive the majority of sales volume, and higher-priced items sell in moderately lower quantities. Together, these findings provide a coherent picture of the retailer's sales dynamics and offer a practical foundation for decisions related to inventory management, international market development, promotional planning, and product portfolio strategy.

Future work could extend this analysis in several directions: a customer segmentation model using RFM (Recency, Frequency, Monetary) analysis, a time-series forecasting model to predict future monthly revenue, or a product recommendation system based on co-purchase patterns. Incorporating more recent data and additional customer attributes would substantially increase the analytical depth and commercial applicability of these extensions.

12. References / Dataset Source

1. Chen, D. (2015). Online Retail Dataset. UCI Machine Learning Repository. Available at: https://www.kaggle.com/datasets/lakshmi25npathi/online-retail-dataset

13. Appendix (Optional)

This section may be used to include additional supporting material such as:

- Additional charts or exploratory plots not included in the main body
- Extended summary statistics tables
- Code screenshots or Jupyter Notebook cell outputs
- Expanded outlier listings or data quality reports